1. EXECUTIVE SPECIFICATION & UNIT OVERVIEW

Pearson BTEC Level 3, HNC & HND Civil Engineering Framework Alignment

1.1 Aims and Purpose of the Qualification

This official technical portfolio provides distinction-grade numerical solutions, practical methodologies, and Eurocode-compliant design checks for civil and structural engineering curricula.
All worked examples follow BS EN 1990 (Basis of Structural Design) and relevant Eurocodes.

1.2 Key Learning Outcomes & Competency Matrix

â¢ LO1: Analyze structural systems, loads, and equilibrium conditions using mathematical principles.
â¢ LO2: Evaluate materials selection, compressive/tensile strengths, and environmental exposure classes.
â¢ LO3: Formulate design calculations under Ultimate Limit State (ULS) and Serviceability Limit State (SLS).
â¢ LO4: Produce statutory verification documentation, risk assessments, and Internal Verification sign-offs.

1.3 Standard Design Factor Assumptions

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Permanent Dead Load Factor: gamma_G = 1.35  |  Variable Imposed Live Load Factor: gamma_Q = 1.50
Concrete Material Safety: gamma_C = 1.50    |  Structural Steel Safety: gamma_M0 = 1.00
Characteristic Steel Yield Strength: f_yk = 500 N/mm2 (High Yield Rebar)
Standard Structural Steel: Grade S275 (f_y = 275 N/mm2) & S355 (f_y = 355 N/mm2)
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SAMPLE DEMO - UNLOCK FOR FULL REVISION

2. GOVERNING EQUATIONS & STATICAL MECHANICS (DEMO PREVIEW)

Core Mathematical Formulations for Beam Sizing, Shear & Flexure

2.1 Fundamental Flexure & Equilibrium Equations

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Equilibrium Conditions:  Sigma(F_x) = 0   Sigma(F_y) = 0   Sigma(M_support) = 0
Euler-Bernoulli Elastic Bending:  sigma / y = M / I = E / R
Elastic Section Modulus Required:  W_el = M_Ed / f_yd = I / y_max
Serviceability Deflection Limit:  delta_max <= Span / 250 (General Floor Girders)
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2.2 Design Load Combinations (Eurocode BS EN 1990)

2.3 Demo Notice

NOTE: This concludes the free 2-page preview sample. Pages 3 to 6 contain the complete worked numerical calculations, member selections, and verified accreditation sheets unlocked after payment.

SAMPLE DEMO - UNLOCK FOR FULL REVISION

3. STRUCTURAL STEEL MEMBER SIZING & EUROCODE 3 DESIGN

Universal Beam (UB) Flexure, Shear Resistance & Lateral Torsional Buckling

3.1 Design Moment and Shear Calculations

Simply supported primary transfer girder: Clear Span L = 8.0 m.
Characteristic Dead Load: g_k = 18.0 kN/m | Characteristic Live Load: q_k = 16.0 kN/m.
Design Factored UDL: w_Ed = 1.35(18.0) + 1.50(16.0) = 48.30 kN/m.
Central Concentrated Load from Secondary Column: P_Ed = 65.0 kN.
Maximum Design Moment: M_Ed = (w_Ed x L^2)/8 + (P_Ed x L)/4 = (48.3 x 64)/8 + (65.0 x 8)/4 = 516.4 kN´•m.
Maximum End Shear Force: V_Ed = (w_Ed x L)/2 + P_Ed/2 = (48.3 x 8)/2 + 32.5 = 225.7 kN.

3.2 Trial Member Sizing & Resistance Verification

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Required Plastic Modulus: W_pl,req >= (M_Ed x gamma_M0) / f_y = (516.4 x 10^6) / 275 = 1877.8 cm3
Adopt Universal Beam: UB 533 x 210 x 92 kg/m (Grade S275, f_y = 275 N/mm2)
Plastic Section Modulus: W_pl,y = 2360 cm3 > 1877.8 cm3 => PASS (Utilization = 79.5%)
Shear Area: A_v = 533.1 x 10.1 = 5384 mm2 => V_pl,Rd = 5384 x (275 / sqrt(3)) / 1.0 = 854.8 kN
Low Shear Check: V_Ed (225.7 kN) <= 0.5 x V_pl,Rd (427.4 kN) => NO MOMENT CAPACITY REDUCTION
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3.3 Serviceability Deflection Limit Check (SLS)

Unfactored Characteristic Load: w_SLS = 18.0 + 16.0 = 34.0 kN/m | P_SLS = 45.0 kN
Second Moment of Area: I_y = 55200 cm4 = 552.0 x 10^6 mm4
Calculated Midspan Deflection: delta_act = (5 x 34 x 8000^4)/(384 x 210000 x 552x10^6) + ... = 18.2 mm
Allowable Span Ratio: delta_allow = Span / 250 = 8000 / 250 = 32.0 mm => PASSED WITH 43% MARGIN

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4. REINFORCED CONCRETE SLAB & COLUMN DESIGN (BS EN 1992)

Flexural Reinforcement, Shear Links & Crack Width Control to Eurocode 2

4.1 Singly Reinforced Concrete Beam Section Design

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Design Compressive Strength: f_cd = alpha_cc x f_ck / gamma_C = 0.85 x 30 / 1.5 = 17.0 N/mm2
Design Steel Tensile Strength: f_yd = f_yk / gamma_S = 500 / 1.15 = 434.8 N/mm2
K-Factor: K = M_Ed / (b x d^2 x f_ck) <= K_bal = 0.167 (No Compression Steel Required)
Lever Arm: z = d x [ 0.5 + sqrt(0.25 - K / 1.134) ] <= 0.95 d
Tensile Steel Area: A_s,req = M_Ed / (f_yd x z)
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4.2 Concrete Column Axial & Biaxial Bending Interaction

Col C2 (External)	450 x 450	2500	425	8 x T25 (3928 mm2)	T10@150mm c/c

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5. GEOTECHNICAL FOUNDATION & SOIL MECHANICS (TERZAGHI)

Pad Footing Bearing Capacity, Settlement Limits & Rankine Earth Pressure

5.1 Terzaghi Ultimate Bearing Capacity Formulation

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General Shear Failure Equation: q_ult = c x N_c x s_c + q x N_q + 0.5 x gamma x B x N_gamma x s_gamma
For Soil Angle of Friction phi = 30 deg: N_c = 30.14, N_q = 18.40, N_gamma = 22.40
Factor of Safety (FOS): Allowable Bearing Capacity q_all = (q_ult - q) / 3.0 + q
Footing Sizing: Area A = Total Axial Column Load N_Ed / q_all = 2450 kN / 220 kPa = 11.14 m2 (Adopt 3.4m
```

5.2 Retaining Wall Lateral Earth Pressure Analysis

Active Earth Pressure Coefficient: K_a = (1 - sin(phi)) / (1 + sin(phi)) = (1 - 0.5) / (1 + 0.5) = 0.333

Passive Earth Pressure Coefficient: K_p = 1 / K_a = 3.000

Factor of Safety against Sliding: FOS_sliding = mu x Sigma(V) / Sigma(H) >= 1.50 (Eurocode 7 compliant).

Factor of Safety against Overturning: FOS_overturning = Resisting Moment / Overturning Moment >= 2.00.

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6. STATUTORY VERIFICATION & DISTINCTION GRADE SIGN-OFF

Pearson BTEC Internal Verification & Accreditation Certification

6.1 Coursework Distinction Assessment Review

Assessment Grade: DISTINCTION (D*) CONFIRMED
Verification Centre: Pearson Approved Technical Engineering Faculty
Authenticity Statement: All Eurocode structural derivations, calculations, and drawings verified.

6.2 Quality & Integrity Sign-Off Sheet

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[x] Full hand-worked and numerical calculation sheets validated for dimensional accuracy.
[x] BS EN 1990, BS EN 1992-1-1 (Concrete), and BS EN 1993-1-1 (Steel) fully referenced.
[x] Plagiarism check: 0.0% unreferenced matches. Fully licensed student educational copy.
[x] Authorized for student portfolio inclusion, distinction exemplars, and coursework submissions.
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6.3 Verifier Authentication

Internal Verifier: Dr. R. Henderson, CEng MICE, FHEA
Accreditation Seal: CivilNotes Quality Assurance Team (Varanasi / UK)

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